

6. Project Testing Phase

ComicCraft – AI Comic Story Creator Using Gemini Models

Verifying the generation workflow, routes, interface, and export

1. Testing Objective

Testing verifies whether the integrated ComicCraft application can receive user inputs, generate comic content, show the generated result, and export it as a PDF. The provided guide focuses on local testing and verification and does not include a formal test report or measured test results.

2. Environment and Server Check

Make sure the virtual environment is active and all required packages are installed. Start the application from the project root using `uvicorn app.main:app --reload` and confirm that the local server runs successfully.

3. Homepage and Input Checks

Open `127.0.0.1:8000` and check that the homepage loads correctly. Verify that the story prompt, character name, setting, tone, and art-style fields are displayed and can be submitted.

4. Form-Based Generation Check

Submit a sample story through `/generate`. Confirm that the route receives the form values and calls the outline, story, image, layout, and PDF functions in the documented order.

5. JSON API Check

Open `/docs` and test `/generate-comic/json` using a JSON request that follows the `PromptRequest` schema. Verify that the endpoint provides the documented layout information and PDF path.

6. Image Generation Check

Use `/test-image` to independently test image generation with a direct prompt. Confirm that an image is created and saved inside the `static/panels` directory.

7. Preview Check

Review `comic_preview.html` and verify that panels are shown in sequence with their titles, illustrations, scene descriptions, captions, and narration. Check that every image matches its respective panel.

8. PDF Export Check

Use the download option on the preview page. Verify that the PDF is created with panel images and narrative text on its pages, and confirm that the application shows the export-success page.

9. Error and Configuration Checks

Verify that missing or invalid request data is handled through validation and that unexpected problems are managed using the documented exception mechanisms. Confirm that API credentials are provided through environment variables and are not exposed in the page content.

10. Testing Outcome

The testing phase should document pass/fail observations for each check, together with any identified defects and their retest results. The guide explains the testing workflow, but it does not claim that all listed tests have actually been executed or passed.